

# A Novel Browser-based No-code Machine Learning Application Development Tool

Erol Ozan, Ph.D.  
Department of Technology Systems  
East Carolina University

Greenville, North Carolina, USA  
[ozang@ecu.edu](mailto:ozang@ecu.edu)

**Abstract**—Machine learning (ML) application development constitutes a complex process that requires developers to have considerable amount of programming and specialized technical skills. There is still a substantial barrier for researchers who do not possess the programming and technical skills to develop ML models that meets the needs of their specific fields of study. The typical ML model development workflow entails the installation of a set of software elements and a certain amount of code development. This paper introduces a browser-based ML application development tool that is geared towards the needs of researchers who have limited knowledge in programming. The tool allows the users to create a customized image classifier model based on the image sets that they provide. The tool provides a no-code workflow that enables users to create and test their ML models on a browser without downloading any software modules.

**Keywords**—Machine learning, artificial intelligence application development tools, browser-based machine learning systems, system design, machine learning model development

## I. INTRODUCTION

There is considerable interest in building easy to use artificial intelligence (AI) application development tools that can be used by individuals with various background. The objective of such tools is to make the AI technology accessible for the wider segments of potential users. There are significant benefits in incorporating AI technology in various domains of the economy. Faster rates of AI technology adoption depend on the effectiveness of the development tools offered by the AI community. Recently, the variety and capability of no-code ML application development tools demonstrate a high pace of improvement and optimization. The notable examples of such no-code tools are *Teachable Machine* by Google [1], *Obviously.ai* [2], and *Runway* [3].

This paper introduces a new no-code ML application development tool [4] that is light-weight and simple to use. The tool focuses on one category of ML: image classification. In this paper, the main features of the tool set are explained together with the ML model development workflow.

## II. MAIN COMPONENTS OF THE TOOL SET

The tool described in this paper was based on TensorFlowJS platform. The code builds on the open source provided by Google in [5]. The tool features three main elements of

functionality that facilitate the following tasks: (1) preparation of a coherent image set for the ML model creation; (2) training of the ML model files with the custom image set; and (3) running the model with the input image selected by the user.

### A. Image Set Preparation Tool

This function allows users to select the images that will be used in the model training. The user selects the folders that contain the image sets one by one, each folder containing the images representing a distinct class. Each folder has to contain same number of images per distinct class. The number of folders and the number of classes should be equal. The user interface contains textboxes where the user provides other relevant input including the following: total number of images; number of training images; number of test images; number of classes. Fig.1 shows the user interface elements.

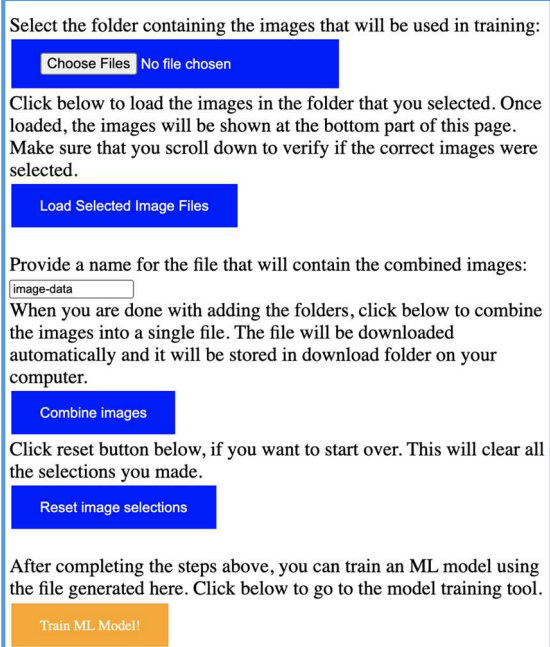


Fig.1. Screenshot showing the image set preparation tool's user interface.

Table 1. Model Architecture

| Layer                       | Layer Parameters                                                                                                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| conv2d_Conv2D1              | inputShape: [IMAGE_WIDTH, IMAGE_HEIGHT, IMAGE_CHANNELS],<br>kernelSize: 5,<br>filters: 8,<br>strides: 1,<br>activation: 'relu',<br>kernelInitializer: 'varianceScaling' |
| max_pooling2d_MaxPooling2D1 | poolSize: [2, 2], strides: [2, 2]                                                                                                                                       |
| conv2d_Conv2D2              | kernelSize: 5,<br>filters: 16,<br>strides: 1,<br>activation: 'relu',<br>kernelInitializer: 'varianceScaling'                                                            |
| max_pooling2d_MaxPooling2D2 | poolSize: [2, 2], strides: [2, 2]                                                                                                                                       |
| flatten_Flatten1            |                                                                                                                                                                         |
| dense_Dense1                | units: NUM_OUTPUT_CLASSES,<br>kernelInitializer: 'varianceScaling',<br>activation: 'softmax'                                                                            |

The number of images in the data set:  
160

The number of training images in the data set:  
160

The number of test images in the data set:  
160

The resulting image size in pixels:  
40

The number of classes:  
3

number\_of\_images\_in\_the\_class:  
100

Enter the class names seperated with commas (example: classname1,classname2,classname3...):  
class1,class2

Select an Image Block File to Load:  
Choose File No file chosen

Load Selected File

Enter batch size:  
10

Enter a value for epochs:  
10

Provide a name for the model:  
my-model

Train the model

Fig.2. Screenshot showing the model trainer's user interface.

### B. Model Trainer

This function enables users to train a ML model by selecting the ML model and adjusting the training parameters. The output of this function is comprised of two files: model architecture file and the model weights file. Those two files are sufficient to create the applications that can be integrated into a web application. The program uses the architecture featured in the

tutorial provide by Google [5]. The key parameters of the model architecture are shown in Table 1. The description of the parameters are as follows. The convolution filter layer (conv2d\_Conv2D1) uses a convolutional window, which is a square matrix. The size of the matrix is determined by the parameter *kernelSize*. The parameter, *filters* represents the number of filters that will be applied to the input data. The term, *strides* refers to the size of the window (in pixels), which slides over the image data matrix. A strides value of 1 means that the filter would slide by 1 pixel for each step. The model architecture uses *Rectified Linear Unit (relu)* function, which is commonly used in many ML applications. The parameter *kernelInitializer* identifies the method used for initializing the model weights. *VarianceScaling* was selected as the method to initialize the weights [5]. Fig.2 shows the user interface elements.

### C. Image Class Prediction Generator

This function provides a capability to run the selected model with the input image selected by the user. The ML model is loaded into the web page via the use of a method in TensorFlowJS shown in the fig.3 below.

```
const model_loaded = await tf.loadLayersModel(modelfilename);
```

Fig.3. Code snippet for loading an ML model into the page

The predictions of the model are displayed in terms of probability values for each class. Fig.4 shows the code snippet used to generate the class predictions for the input image.

```
var preds = model_loaded.predict(input_image);  
preds.print();
```

Fig.4. Code snippet for outputting model predictions

A screenshot of the tool output for the class predictions is shown in fig. 5.

## III. DESCRIPTION OF THE WORKFLOW

This section gives an overview of the steps involved in building a ML model using the tool set. At the first stage, the user creates a set of folders and stores the images into the folders according to the classes. Next, the user loads the images into the tool via the user interface. The tool outputs a text file that combines the content of the all the images selected by the user. The output file contains the numeric values of the pixels of each image as a combined array. The format of the file is a simple text file. The following stage is training the ML model. At that step user loads the output file that combines the images into the model trainer. The user interface enables users to adjust batch size and epochs according to the specific characteristics of the project. The output of the model training module constitute two files with extensions ".json" and "weights.bin."

#### IV. SYSTEMS REQUIREMENTS

The tool is designed to run on a computer that has an active server running locally. A simple local http server such as *Web Server for Chrome* [6] is sufficient for the purpose of using the tool. It is recommended that all the files and folders to be stored in the folder that is designated as the root directory of the server.

The tool was tested on a Chrome browser (Version 89.0.4389.82; official build; x86\_64) running on a MacBook Pro 14,1 with an Intel Core i7 processor (2.5 GHz). The computer's installed memory was 16 GB.

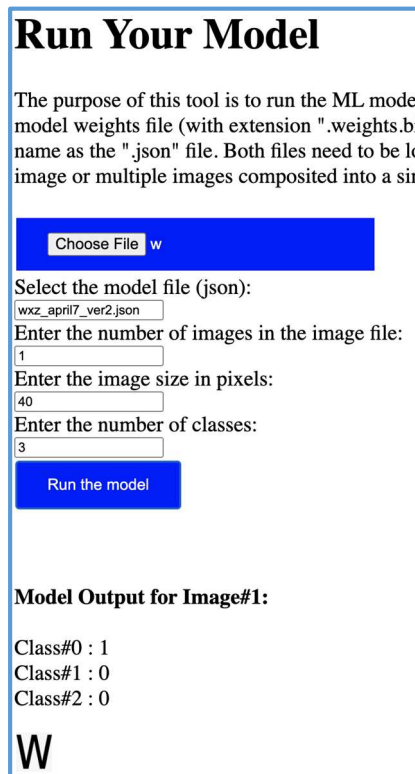


Fig.5. Screenshot showing the model predictions generated by the tool.

#### V. CONCLUSION

This paper reported the current status of an open-source software development project. The main features of the tool set were described. There were certain features that were still being built at the time of writing of this paper. The future versions of the tool will provide a capability to edit the model architecture. The efforts will also focus on optimizing the file structure that combines the images provided by the user.

#### REFERENCES

- [1] "Teachable Machine," Google, 2021. [Online]. Available: <https://teachablemachine.withgoogle.com/>. [Accessed 10 April 2021].
- [2] Obviously.ai, 2021. [Online]. Available: <http://obviously.ai>. [Accessed 10 April 2021].
- [3] "Runway ML," Runway Ai Inc., 2021. [Online]. Available: <http://www.runwayai.com>. [Accessed 30 April 2021].
- [4] E. Ozan, "ML\_Tool," 2021. [Online]. Available: [https://github.com/ozanix/ml\\_tool](https://github.com/ozanix/ml_tool). [Accessed 10 April 2021].
- [5] "TensorFlow.js — Handwritten digit recognition with CNNs," Google, 2021. [Online]. Available: <https://codelabs.developers.google.com/codelabs/tfjs-training-classification/index.html#4>. [Accessed 10 April 2021].
- [6] "Web Server for Chrome," Chromebeat.com, 2021. [Online]. Available: <https://chrome.google.com/webstore/detail/web-server-for-chrome>. [Accessed 10 April 2021].